

B2 Adding Vectors

Where bearings are given, they are in degrees East of North (so North is 000° , East is 090° , South is 180° and West 270°). For the purposes of this exercise, assume that the Earth is flat.

- B2.1 How far am I from my starting point if I cycle 12 km West and then 8.0 km South?
- B2.2 One day, the tide in a river is running East at 0.8 m s^{-1} . A swimmer faces North and swims, relative to the water, at 0.6 m s^{-1} .
- How fast does the swimmer move, as viewed by a hovering wasp?
 - In which direction does the wasp see the swimmer move? Give your answer in degrees as a bearing.
- B2.3 One day, the wind is blowing from the west at 35 km h^{-1} . A pilot is in charge of an aeroplane which can fly at 70 km h^{-1} in stationary air. The pilot wants to fly South.
- In which direction should the pilot point the aircraft? Give your answer in degrees as a bearing.
 - How fast will the aircraft fly over the ground?

CHAPTER B. MECHANICS

Answers

B2.1 $\sqrt{12^2 + 8^2} = 14.42 \text{ km}$

B2.2

a) $\sqrt{0.6^2 + 0.8^2} = 1.0 \text{ m s}^{-1}$

b) $\arctan\left(\frac{0.8}{0.6}\right) = 053^\circ$

B2.3

a) Angle west of south = $\arcsin\left(\frac{35}{70}\right) = 30^\circ$.
Bearing = $180 + 30 = 210^\circ$

b) $\sqrt{70^2 - 35^2} = 16.8 \text{ km h}^{-1}$